

For decades, sickle cell disease (SCD) treatments focused primarily on managing symptoms and diluting sickled cells. However, with its approval by the FDA in 2023, Casgevy (the first CRISPR-Cas9-based therapy) revolutionized gene-based medicine by offering a definitive approach to treating SCD for patients 12 years or older [10]. A patient's own blood-forming stem cells are reprogrammed to increase fetal hemoglobin (HbF) levels and prevent sickling, which confers several secondary advantages, such as eliminating the need for recurrent transfusions/transplants [3]. Despite these benefits, ethical dilemmas and clinical risks present challenges, raising questions about whether Casgevy's potential outweighs its uncertainties and whether one would choose this treatment.

Casgevy works by restoring production of fetal hemoglobin (HbF), a protective form of hemoglobin made before birth that prevents red blood cells from sickling. However, after birth, BCL11A turns off HbF production [10]. To keep HbF levels elevated, Casgevy utilizes CRISPR-Cas9, a genome-editing tool with two components: the Cas9 protein that cuts the DNA, and a single guide RNA that guides Cas9 to the specific location in the DNA to edit. It modifies the enhancer region of the BCL11A gene, effectively "breaking the switch" and turning on production of HbF. The process begins with extraction, where the patient's stem cells are "mobilized" into the bloodstream and collected through apheresis, then sent to a manufacturing facility for CRISPR-Cas9 editing. While editing, the patient undergoes conditioning, which involves intensive myeloablative chemotherapy to eliminate any remaining stem cells to make space for edited ones [3]. Once editing is finished, a one-time procedure reinfuses the cells, allowing the cells to engraft (take hold) in the bone marrow and begin producing cells that produce fetal hemoglobin [4].

Casgevy offers several advantages, most notably its definitive approach to gene editing that can potentially cure, or at least eliminate vasoocclusive crises, painful blockages of blood flow formed when red blood cells sickle [4]. Unlike traditional therapies such as hydroxyurea or chronic blood transfusions, Casgevy aims for a lasting cure [10]. Additionally, it provides a comparative advantage over conventional bone marrow transplant treatments because it doesn't require a matched donor, a barrier for ~80-85% of patients [4]. Donor transplants are also often accompanied by lifelong immunosuppressive drugs, which Casgevy effectively avoids.

There is proven clinical success of Casgevy in SCD treatment. For instance, one particular study found that 36 out of 39 (92.3%) patients with at least 16 months of follow-up were free from vasoocclusive crises for at least 12 consecutive months [6]. Another trial found that 29 out of the 30 adults and adolescents were free of severe pain crises for 22.2 months [3].

Despite its potential, Casgevy brings many drawbacks and risks. One such disadvantage is the side effects of conditioning chemotherapy. The removal of stem cells from the bone marrow for editing is often described as an excruciatingly painful process, with patients facing severe side effects like painful mouth sores, vomiting, and hair loss [8]. The reduced white blood cell count also leaves the patient vulnerable to infections [4]. Considering that Casgevy is relatively new, information regarding long-term side effects and durability is relatively unknown. Additionally, off-target effects may occur when Cas9 edits unintended sections of DNA, potentially causing detrimental mutations [1]. For example, one patient developed an ASXL1 gene mutation after treatment [7].

Casgevy's listing price of \$2.2 million per patient positions it as one of the most expensive treatments [9], presenting a concerning ethical contradiction, as areas that require the treatment the most are least likely to access the treatment due to economic barriers. Research shows that 80% of sickle cell patients live in Africa, yet basic SCD treatments like hydroxyurea are out of reach for most, let alone Casgevy [2]. Without regulation to ensure distribution, the approval of CRISPR will deepen health inequalities. There are also broader societal concerns. The disability community emphasized that genome editing encourages purging "unacceptable" genetic conditions, framing differences as needing to be fixed [5].

If someone I care about had SCD, I would consider many factors to determine suitability. Even with high costs, its characteristic one-time treatment and success in clinical trials make it compelling. However, the treatment is physically demanding, and the patient risks severe temporary side effects and a permanent risk of infertility [3]. Long-term outcomes are also uncertain. So, for older patients with accumulated organ damage or those whose disease is less severe, the risks may outweigh the advantages. But for the large portion of patients who lack a matched bone marrow donor, Casgevy may be suitable.

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